

Low-Cost Smart Agriculture System

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This IoT-based project monitors soil moisture, temperature, and humidity levels near plants and light intensity near the control station. It turns on water solenoid valve when moisture level goes below the threshold level. Its vibration sensor detects movement or any other unwanted activity of animals and rodents near the plantation area.

The project is equally suitable for indoor plants in buildings, greenhouse

farming, and for regular farming. Adafruit IoT platform is used to monitor various sensors' status online.

Block diagram of the low-cost smart agriculture system is shown in Fig. 1. The author's prototype of the project is shown in Fig. 2 while its circuit diagram is shown in Fig. 3.

Circuit and working

The heart of the IoT system is the NodeMCU ESP-32 8266 Wi-Fi module.

It is used to transmit the signals collected from various sensors to the users through a computer network. The sensors include four soil moisture sensors (SS1 through SS4), a light-dependent resistor (LDR), a vibration sensor (VS1), and a temperature and humidity sensor (DHT11).

If soil moisture or humidity level near any plant(s) goes below the threshold level (as per the program), the controller in NodeMCU switches

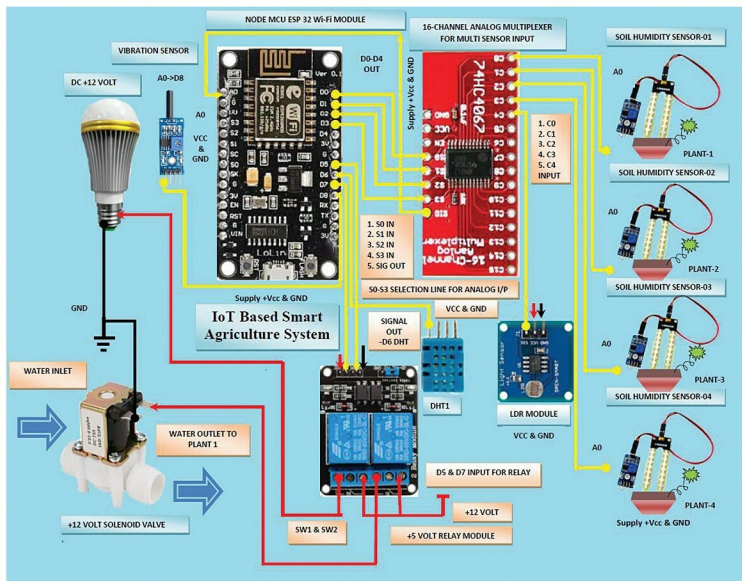


Fig. 1: Block diagram of the smart agriculture system